

- 4 Applicants for a particular college take a written test when they attend for interview. There are two different written tests, A and B , and each applicant takes one or the other. The interviewer wants to determine whether the medians of the distribution of marks obtained in the two tests are equal. The marks obtained by a random sample of 8 applicants who took test A and a random sample of 8 applicants who took test B are as follows.

Test A	46	32	29	12	33	18	25	40
Test B	36	28	49	37	48	35	41	31

- (a) Carry out a Wilcoxon rank-sum test at the 5% significance level to determine whether there is a difference in the population median marks obtained in the two tests. [6]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

The interviewer considers using the given information to carry out a paired sample t -test to determine whether there is a difference in the population means for the two tests.

- (b) Give two reasons why it is not appropriate to use this test. [2]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dashed lines, creating a series of uniform gaps for writing. The lines are evenly spaced across the entire page, providing a guide for letter height and placement. There is no text or other markings on the page.